

|                                              |                                 |
|----------------------------------------------|---------------------------------|
| Program : <b>Diploma in Microelectronics</b> |                                 |
| Course Code: <b>5499A</b>                    | Course Title: <b>VHDL Lab</b>   |
| Semester: <b>5</b>                           | Credits: <b>1.5</b>             |
| Course Category: <b>Program Elective</b>     |                                 |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>   | Periods per semester: <b>45</b> |

### Course Objectives:

- To simulate and test the functional aspects of digital electronic circuits using the available software tool (eg: Xilinx ISE, Xilinx Vivado, Modelsim etc.)

### Course Prerequisites:

| Topic                               | Course code | Course name         | Semester |
|-------------------------------------|-------------|---------------------|----------|
| Combinational & Sequential circuits |             | Digital Electronics | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                   | Duration (Hours) | Cognitive level |
|-----------------|-------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Develop simulation models to implement various logic gates                    | 6                | Applying        |
| CO2             | Develop simulation models to implement various combinational logic circuits   | 12               | Applying        |
| CO3             | Develop simulation models to implement various flip flops and shift registers | 9                | Applying        |
| CO4             | Develop simulation models to implement various counters                       | 15               | Applying        |
|                 | Lab Exam                                                                      | 3                |                 |

**CO-PO Mapping:**

| <b>Course Outcomes</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| <b>CO1</b>             | 3          | 3          | 3          |            |            |            |            |
| <b>CO2</b>             | 3          | 3          | 3          |            |            |            |            |
| <b>CO3</b>             | 3          | 3          | 3          |            |            |            |            |
| <b>CO4</b>             | 3          | 3          | 3          |            |            |            |            |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b> | <b>Description</b>                                                                                                | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>Develop simulation models to implement various logic gates</b>                                                 |                         |                        |
| M1.01                  | Simulate and verify the truth tables of logic gates – NOT, OR, AND, NAND, NOR, XOR XNOR                           | 3                       | Applying               |
| M1.02                  | Develop a simulation model of basic gates using universal gates and verify the functionality                      | 3                       | Applying               |
| <b>CO2</b>             | <b>Develop simulation models to implement various combinational logic circuits</b>                                |                         |                        |
| M2.01                  | Develop a simulation model of Half Adder and Full Adder using dataflow modeling and structural modeling           | 3                       | Applying               |
| M2.02                  | Develop a simulation model of Half Subtractor and Full Subtractor using dataflow modeling and structural modeling | 3                       | Applying               |
| M2.03                  | Develop a simulation model of 4-bit ripple carry adder using structural modeling                                  | 3                       | Applying               |
| M2.04                  | Develop a simulation model of 4:1 Multiplexer, 1:4 De-Multiplexer and code converters                             | 3                       | Applying               |
|                        | Lab Exam                                                                                                          | 1.5                     |                        |
| <b>CO3</b>             | <b>Develop simulation models to implement various flip flops and shift registers</b>                              |                         |                        |
| M3.01                  | Develop a simulation model of D, JK and T Flip Flops using behavioral modeling                                    | 3                       | Applying               |

|            |                                                                                           |     |          |
|------------|-------------------------------------------------------------------------------------------|-----|----------|
| M3.02      | Develop a simulation model of D, JK and T Flip Flops using structural modeling            | 3   | Applying |
| M3.03      | Develop a simulation model of shift registers                                             | 3   | Applying |
| <b>CO4</b> | <b>Develop simulation models to implement various counters</b>                            |     |          |
| M4.01      | Develop simulation models to implement Asynchronous 4-bit Upcounter and 4-bit Downcounter | 3   | Applying |
| M4.02      | Develop simulation models to implement Synchronous 4-bit Upcounter and 4-bit Downcounter  | 3   | Applying |
| M4.03      | Develop simulation models to implement Ring counter and Johnson counter                   | 3   | Applying |
| M4.04      | Develop simulation models to implement Asynchronous Mod-5, Mod-10 and Mod-12 counters     | 6   | Applying |
|            | Lab Exam                                                                                  | 1.5 |          |

### Student Activity

#### Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 2-3. There is no duplication allowed in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should include the open ended experiment in their lab record.

Examples:

1. ALU
2. Counters etc.

#### Text /Reference:

| T/R | Book Title/Author                                                                        |
|-----|------------------------------------------------------------------------------------------|
| T1  | VHDL Primer Third edition by J. Bhaskar, PHI                                             |
| R1  | Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, 2/e, TMH. 2008 |
| R2  | Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008.                                |
| R3  | Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008                      |
| R4  | Pedroni V. A., Circuit design with VHDL, PHI, 2008.                                      |
| R5  | Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996                       |

**Online Resources:**

| <b>Sl. No</b> | <b>Website Link</b>                                                                                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://www.xilinx.com/support/documents/sw_manuals/xilinx2022_1/ug900-vivado-logic-simulation.pdf">https://www.xilinx.com/support/documents/sw_manuals/xilinx2022_1/ug900-vivado-logic-simulation.pdf</a>   |
| 2             | <a href="https://www.xilinx.com/support/documentation-navigation/self-paced-tutorials/see-all-tutorials.html">https://www.xilinx.com/support/documentation-navigation/self-paced-tutorials/see-all-tutorials.html</a> |
| 3             | <a href="https://www.eecs.umich.edu/courses/doing_dsp/handout/vhdl-tutorial.pdf">https://www.eecs.umich.edu/courses/doing_dsp/handout/vhdl-tutorial.pdf</a>                                                           |
| 4             | <a href="https://www.ee.iitb.ac.in/~smdp/DKStutorials/vhdl-overview.pdf">https://www.ee.iitb.ac.in/~smdp/DKStutorials/vhdl-overview.pdf</a>                                                                           |
| 5             | <a href="https://www.cdeep.iitb.ac.in/slides/A09/EE705/EE705-L19.pdf">https://www.cdeep.iitb.ac.in/slides/A09/EE705/EE705-L19.pdf</a>                                                                                 |